

Worksheet No. - 4

Student Name: Anjali kumari

Branch: MCA general

Semester: 3rd

Subject Name: MACHINE LEARNING.

Subject Code: CONT_25CAP-702

UID: 25MCA20023

Section: Group:1/A

Date of Performance: 12th sept.

Experiment 4: Implement classification algorithms—Decision Tree on a suitable dataset and evaluate their performance using confusion matrix, precision, recall, and F1-score.

1.Aim:

To develop a **Decision Tree Classification model** for predicting the likelihood of stroke using patient health and demographic information and to evaluate its performance using various classification metrics.

2.Objective:

- To preprocess and clean the stroke prediction dataset.
- To identify suitable features for predicting stroke.
- To train a Decision Tree Classifier on the dataset.
- To evaluate the model using **Accuracy, Precision, F1-score, and Confusion Matrix**.
- To analyze the effect of different **maximum tree depths** on model performance.
- To identify **underfitting and overfitting** by comparing training and testing accuracy.
- To select an appropriate tree depth based on testing performance.

3.Task to be done :

- Load and preprocess the stroke dataset.
- Train a Decision Tree Classifier.
- Evaluate the model using Accuracy, Precision and F1-score.
- Plot the Confusion Matrix.
- Analyze different tree depths for underfitting and overfitting.

4. Online Resources:

1. NumPy Documentation
2. Pandas Documentation
3. Scikit-Learn Official Documentation
4. Matplotlib Official Documentation
5. Seaborn Documentation
6. Kaggle Titanic Dataset Resources

Software Requirements

1. Python 3.x
2. Jupyter Notebook / Google Colab
3. Anaconda Distribution

Standard Tools Used

1. NumPy
2. Pandas
3. Matplotlib
4. Seaborn
5. SciPy
6. Scikit-Learn

5. Procedure/ Algorithm:

```
# =====
```

```
# STROKE PREDICTION USING DECISION TREE CLASSIFIER
```

```
# =====
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
```

```
from sklearn.metrics import (
```

```
    accuracy_score,
```

```
    precision_score,
```

```
    f1_score,
```

```
    confusion_matrix,
```

```
    classification_report,
```

```
    ConfusionMatrixDisplay
```

```
)
```

```
# =====
```

```
# 1. LOAD DATASET
```

```
# =====
```

```
file_path = r"C:\Users\Dell\Downloads\archive (1)\healthcare-dataset-stroke-data.csv"
```

```

print("Dataset Shape:", df.shape)
print("\nFirst 5 Records:")
print(df.head())

# =====
# 2. CHECK MISSING VALUES
# =====

print("\nMissing Values:")
print(df.isnull().sum())

# =====
# 3. REMOVE ID COLUMN
# =====

df = df.drop("id", axis=1)

# =====
# 4. HANDLE MISSING VALUES
# =====

df["bmi"] = df["bmi"].fillna(df["bmi"].median())

# =====
# 5. ENCODE CATEGORICAL VARIABLES
# =====

categorical_columns = [
    "gender",
    "ever_married",
    "work_type",
    "Residence_type",
    "smoking_status"
]

for column in categorical_columns:
    encoder = LabelEncoder()
    df[column] = encoder.fit_transform(df[column])

# =====
# 6. SEPARATE FEATURES AND TARGET
# =====

X = df.drop("stroke", axis=1)
y = df["stroke"]

# =====
# 7. TRAIN TEST SPLIT
# =====

```

```

=====
MODEL PERFORMANCE
=====
Accuracy : 0.6986
Precision : 0.1075
F1 Score : 0.1866

=====
CLASSIFICATION REPORT
=====

```

	precision	recall	f1-score	support
0	0.98	0.70	0.82	1458
1	0.11	0.71	0.19	75
accuracy			0.70	1533
macro avg	0.54	0.70	0.50	1533
weighted avg	0.94	0.70	0.78	1533

Fig1 : MODEL Performance

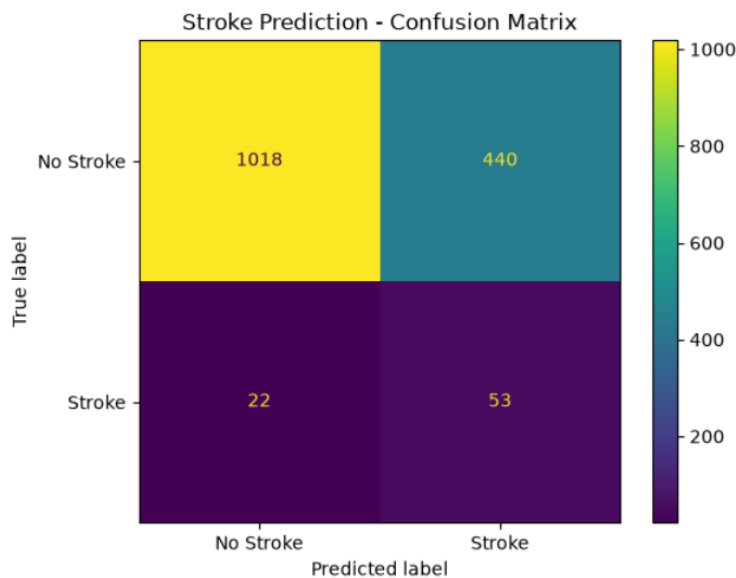


Fig 2 : Confusion Matrix

```

=====
MAX DEPTH ANALYSIS
=====

```

Depth 1	Train Accuracy: 0.5974	Test Accuracy: 0.5864
Depth 2	Train Accuracy: 0.5974	Test Accuracy: 0.5864
Depth 3	Train Accuracy: 0.7330	Test Accuracy: 0.7306
Depth 4	Train Accuracy: 0.6872	Test Accuracy: 0.6745
Depth 5	Train Accuracy: 0.7154	Test Accuracy: 0.6986
Depth 6	Train Accuracy: 0.7143	Test Accuracy: 0.7019
Depth 8	Train Accuracy: 0.7621	Test Accuracy: 0.7371
Depth 10	Train Accuracy: 0.8141	Test Accuracy: 0.7795
Depth 15	Train Accuracy: 0.8356	Test Accuracy: 0.7958

```

=====
BEST DEPTH
=====
Best Depth : 15
Best Test Accuracy: 0.7958

```

Fig 3 : MAX depth Analysis

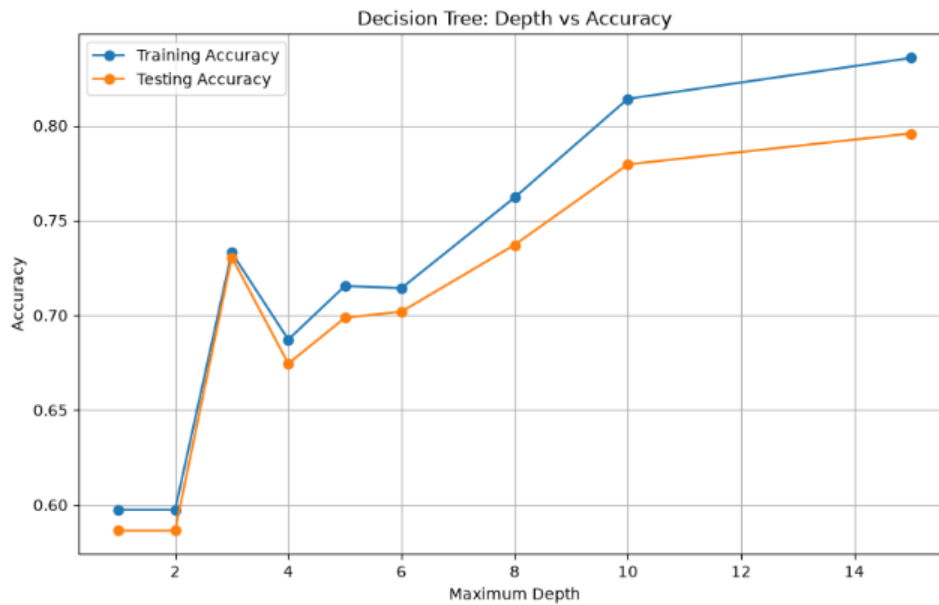


FIG:4 Comparison of Training and Testing of a model.

6.LEARNING OUTCOME:

1. Understand the basic concepts of **Decision Tree Classification**.
2. Perform data preprocessing and categorical data encoding.
3. Divide a dataset into **training and testing sets**.
4. Build and train a Decision Tree model using Python and Scikit-learn.
5. Evaluate classification models using **Accuracy, Precision, Recall, F1-score, and Confusion Matrix**.
6. Understand the impact of model complexity and **maximum tree depth**.
7. Identify **underfitting and overfitting** from training and testing performance.
8. Interpret the results of a machine-learning classification model